Name : D Hari Charan

R.No : 192311319

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Question 1

A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. How many full-duplex voice channels can this spectrum support?

EXPLANATION
A full-duplex channel needs two simplex channels (forward + reverse), so one duplex channel = 2 × 25 kHz = 50 kHz. Number of channels = Total bandwidth ÷ Duplex channel bandwidth = 45 MHz ÷ 50 kHz = 45000 ÷ 50.

ANSWER: 900 full-duplex channels

Question 2

A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. How many channels is a single cell typically allocated?

EXPLANATION
Channels per cell = Total channels ÷ Cluster size N = 560 ÷ 7.

ANSWER: 80 channels per cell

Question 3

A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels.

EXPLANATION
Each carrier carries 8 time slots, and each slot is one voice channel. Total = Carriers × Slots per carrier = 100 × 8.

ANSWER: 800 simultaneous voice channels

Question 4

The base station radiates 80 W while a nearby handset transmits just 4 W. Express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

EXPLANATION
Ratio = 80 ÷ 4 = 20. Ratio in dB = 10·log10(20) = 10 × 1.301.

ANSWER: 13.01 dB

Question 5

A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. Find the C/I ratio in dB.

EXPLANATION
When both values are in dBm, the ratio in dB is the difference: C/I = C(dBm) − I(dBm) = −60 − (−92).

ANSWER: 32 dB

Question 6

An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

EXPLANATION
Total channels = 42 MHz ÷ 300 kHz = 42000 ÷ 300 = 140. Channels per cell = 140 ÷ 7.

ANSWER: 20 channels per cell

Question 7

A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they expect?

EXPLANATION
Received power follows $Pr \propto 1/d^n$. With d doubled and n = 4, the power falls by $2^4 = 16$. New Pr = 64 ÷ 16.

ANSWER: 4 W

Question 8

Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

EXPLANATION
Inverse-square law: $Pr \propto Pt/d^2$. To keep Pr the same when d becomes 2d, Pt must increase by $(2)^2 = 4$.

ANSWER: Scale up by a factor of 4

Question 9

A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the cluster?

EXPLANATION
Total channels = Cluster size × Channels per cell = 4 × 15.

ANSWER: 60 channels

Question 10

A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift.

EXPLANATION
Doppler shift fd = v·f/c. v = 100 km/h = 27.78 m/s; f = 900 × 10^6 Hz; c = 3 × 10^8 m/s. fd = (27.78 × 900×10^6) ÷ (3×10^8).

ANSWER: ≈ 83.3 Hz

Question 11

A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

EXPLANATION
Slot duration = Frame duration ÷ Number of slots = 5 ms ÷ 8.

ANSWER: 0.625 ms

Question 12

Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created?

EXPLANATION
Carriers = Total bandwidth ÷ Spacing = 24 MHz ÷ 200 kHz = 24000 ÷ 200.

ANSWER: 120 carriers

Question 13

A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver sees, in dBm.

EXPLANATION
Convert to dBm: 10·log10(8 W × 1000 mW) = 10·log10(8000) = 39.03 dBm. Received power = Transmit power (dBm) − Path loss (dB) = 39.03 − 33.

ANSWER: ≈ 6.03 dBm

Question 14

A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

EXPLANATION
FSPL (dB) = 32.45 + 20·log10(f in MHz) + 20·log10(d in km). 20·log10(1900) = 65.58; 20·log10(2.5) = 7.96. FSPL = 32.45 + 65.58 + 7.96.

ANSWER: ≈ 106 dB

Question 15

A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

EXPLANATION
Capacity = Cells × Calls per cell = 6 × 250.

ANSWER: 1500 calls

Question 16

A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active at any one time?

EXPLANATION
Active calls = Arrival rate × Average duration (Little's law). Duration = 3 min = 180 s. Active calls = 4000 × 180.

ANSWER: 720,000 calls

Question 17

Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

EXPLANATION
Total bandwidth = Channels × Bandwidth per channel = 150 × 25 kHz = 3750 kHz = 3.75 MHz.

ANSWER: 3.75 MHz

Question 18

Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands. How much total bandwidth is wasted by these guard bands?

EXPLANATION
Using one guard band per band: 30 × 40 kHz = 1200 kHz = 1.2 MHz. (If only the 29 gaps between bands are counted, it would be 29 × 40 = 1.16 MHz.)

ANSWER: 1.2 MHz

Question 19

A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs occur over an hour?

EXPLANATION
Cell diameter = 2 × 1.5 = 3 km. Distance travelled in 1 hour = 90 km. Handoffs = 90 ÷ 3.

ANSWER: 30 handoffs

Question 20

A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the total user-handling capacity?

EXPLANATION
Total users = Centers × Base stations per center × Users per base station = 8 × 40 × 250 = 320 × 250.

ANSWER: 80,000 users

Question 21

Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells are required?

EXPLANATION
R = 400 m = 0.4 km, so cell area = R^2 = 0.16 km^2. Number of cells = Region area ÷ Cell area = 400 ÷ 0.16.

ANSWER: 2500 cells

Question 22

An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out?

EXPLANATION
Channels = Total spectrum ÷ Channel bandwidth = 6000 kHz ÷ 30 kHz.

ANSWER: 200 channels

Question 23

1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

EXPLANATION
Erlang = (Number of calls per hour × Average call duration in minutes) ÷ 60. Total = (1200 × 4 × 2.5) ÷ 60 = 12000 ÷ 60.

ANSWER: 200 Erlangs

Question 24

A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

EXPLANATION
Users = Total bandwidth ÷ Bandwidth per user = 40 MHz ÷ 400 kHz = 40000 ÷ 400.

ANSWER: 100 users

Question 25

A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

EXPLANATION
Total capacity = Users per station × Number of stations = 600 × 1200.

ANSWER: 720,000 users

Question 26

A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

EXPLANATION
Worst case: the subscriber is in the last group paged. Rounds = Total cells ÷ Cells per round = 200 ÷ 4.

ANSWER: 50 paging rounds

Question 27

A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

EXPLANATION
Offered load = Arrival rate × Duration = 40,000 × 3 = 120,000 simultaneous calls. Capacity = 100,000, so excess = 20,000. Blocked fraction = 20,000 ÷ 120,000 = 1/6.

ANSWER: ≈ 16.7 % (1/6) blocked

Question 28

A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

EXPLANATION
Roaming users = 15 % × 600,000 = 0.15 × 600,000.

ANSWER: 90,000 subscribers

Question 29

A switching center is capable of processing 12,000 calls per second, but is currently handling 9,000 calls per second. What percentage of its capacity is utilized?

EXPLANATION
Utilization = (Current load ÷ Capacity) × 100 = (9000 ÷ 12000) × 100.

ANSWER: 75 %

Question 30

An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

EXPLANATION
Total channels = 48 MHz ÷ 200 kHz = 48000 ÷ 200 = 240. Channels per cell = 240 ÷ 4.

ANSWER: 60 channels per cell

Question 31

A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

EXPLANATION
Channels = 30 MHz ÷ 200 kHz = 30000 ÷ 200.

ANSWER: 150 channels

Question 32

Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

EXPLANATION
Successful handoffs = 99 % × 2500 = 0.99 × 2500.

ANSWER: 2475 calls

Question 33

A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

EXPLANATION
Convert to dBm: 10·log10(12000) = 40.79 dBm. Received power = 40.79 − 45.

ANSWER: ≈ −4.21 dBm

Question 34

A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

EXPLANATION
Capacity = Cells × Users per cell = 7 × 900.

ANSWER: 6300 users

Question 35

A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

EXPLANATION
Slot length = Frame duration ÷ Slots = 4.8 ms ÷ 8.

ANSWER: 0.6 ms

Question 36

A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

EXPLANATION
fd = v·f/c. v = 150 km/h = 41.67 m/s; f = 1.8 × 10^9 Hz. fd = (41.67 × 1.8×10^9) ÷ (3×10^8).

ANSWER: 250 Hz

Question 37

A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at 70,000 per minute, each averaging 4 minutes. What share of the offered load gets blocked?

EXPLANATION
Offered load = 70,000 × 4 = 280,000 calls. Capacity = 250,000, so excess = 30,000. Blocked share = 30,000 ÷ 280,000.

ANSWER: ≈ 10.7 % blocked

Question 38

A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

EXPLANATION
Capacity = 9 × 600.

ANSWER: 5400 users

Question 39

A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are dropped every second?

EXPLANATION
Dropped = 0.4 % × 60,000 = 0.004 × 60,000.

ANSWER: 240 calls per second

Question 40

A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

EXPLANATION
Path loss (dB) = Transmitted power (dBm) − Received power (dBm) = 43 − (−87). Antenna gains are ignored.

ANSWER: 130 dB

Question 41

A single base station covers 2.5 km^2 of territory, and the operator plans to deploy 150 identical stations. What total area will the rollout cover?

EXPLANATION
Total area = Area per station × Number of stations = 2.5 × 150.

ANSWER: 375 km^2

Question 42

An operator applies a frequency reuse factor of 6 over a 48 MHz allocation, with each channel taking 200 kHz. How many channels does each cell get?

EXPLANATION
Total channels = 48 MHz ÷ 200 kHz = 240. Channels per cell = 240 ÷ 6.

ANSWER: 40 channels per cell

Question 43

A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

EXPLANATION
Cycles = Total users ÷ Users per cycle = 150,000 ÷ 6000.

ANSWER: 25 cycles

Question 44

A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place?

EXPLANATION
Handoffs = Call duration ÷ Handoff interval = 9 ÷ 3. (Counting only handoffs strictly inside the call would give 2.)

ANSWER: 3 handoffs

Question 45

A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

EXPLANATION
SNR (dB) = Signal (dBm) − Noise (dBm) = −85 − (−115).

ANSWER: 30 dB

Question 46

A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide?

EXPLANATION
Total = Users × Rate per user = 250 × 150 kbps = 37,500 kbps = 37.5 Mbps.

ANSWER: 37.5 Mbps

Question 47

A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

EXPLANATION
Erlang = (Subscribers × Calls per hour × Duration in min) ÷ 60 = (1500 × 4 × 2.5) ÷ 60 = 15000 ÷ 60.

ANSWER: 250 Erlangs

Question 48

A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

EXPLANATION
Total = 250 × 12 kHz = 3000 kHz = 3 MHz.

ANSWER: 3 MHz

Question 49

A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

EXPLANATION
Throughput = 1500 × 600 kbps = 900,000 kbps = 900 Mbps.

ANSWER: 900 Mbps

Question 50

A telecom operator wants each tower to cover a 4 km radius (area = πR^2) and needs to blanket a 350 km^2 service region. Roughly how many towers will the rollout require?

EXPLANATION
Area per tower = $\pi \times 4^2 = 50.27$ km^2. Towers = $350 \div 50.27 = 6.96$, which rounds up to the next whole tower.

ANSWER: 7 towers